Panos Vardanis

Senior MLOps Engineer

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PROFILE

Senior MLOps Engineer with 5+ years of experience designing and operating ML systems in production. Specialized in end-to-end ML lifecycle automation, tooling, and model serving infrastructure across CPU and GPU environments. Track record of delivering measurable performance gains. Strong advocate for sustainable engineering practices including TDD, clean architecture, and cross-team mentoring.

EMPLOYMENT HISTORY

Personal Projects

April 2026 - Present | Berlin, DE

Self-directed engineering work on LLM inference tooling and software architecture

- Building slipstream, a self-hosted LLM inference platform on Kubernetes (Amazon EKS): serving open models on spot GPUs with vLLM, GPU autoscaling on inference signals, spot-eviction survival, cold-start elimination, inference-aware request routing, and prefill/decode disaggregation. Provisioned with Terraform (IaC), benchmarked cost-competitive against commercial LLM APIs on a live scoreboard, with OpenTelemetry observability across Prometheus, Grafana, Sentry, and PostHog
- Building offgrid, an open-source Python CLI/TUI for local LLM inference that pairs any supported runtime with any coding agent and starts the two together in one command - LM Studio behind Claude Code or OpenCode, running entirely on-device
- Completed the ArjanCodes Software Designer Mindset and Software Architect Mindset tracks and the AI Coding Crash Course by Matt Pocock

Senior MLOps Engineer, Wallaroo.AI ↗

February 2023 - June 2025 | Berlin, DE

Wallaroo.AI empowers enterprise AI teams to deploy, observe, test, and optimize ML from the cloud to the edge with efficiency, scale, and ease

- Implemented vLLM with continuous batching and token streaming as a framework in our product by architecting an async/sync bridge for Arrow Flight RPC; collaborated with Qualcomm on their QAIC vLLM fork; benchmarked Qualcomm AI100 vs. NVIDIA across several serving stacks; built the evaluation framework and ran benchmarks - ~1.5s latency / ~550 output tok/s at 4 QPS, ~1550 output tok/s saturation on Llama 3.1 8B / H100
- Architected a Python library that streamlined low-code ML model deployment on Wallaroo platform on CPU & GPU, supporting PyTorch, HuggingFace, Scikit-Learn, XGBoost, ONNX, and custom implementations (BYOP) with minimal user input
- Built an automatic ONNX conversion mechanism that increased native ONNX model coverage on the Rust engine by 60%, achieving up to ~15x faster inference and lower resource consumption
- Contributed Python and Rust integration of hardware-accelerated inference for Ampere, Intel, and Qualcomm, achieving up to 10x speedup
- Contributed to migrating inference protocol from REST to gRPC using Arrow Flight RPC, delivering up to ~3x latency reduction for Audio & CV domains
- Maintained and enhanced a Rust-based model observability component for data drift monitoring on historical and scheduled future data; implemented backwards compatibility between v1 and v2 in the user-facing Python SDK - users migrated via a single flag
- Reduced CI execution, model upload and deployment times ~2x by replacing pip/venv with uv across build and runtime container images, using BuildKit cache mounts and offline wheel installs

Senior MLOps Engineer, Utopia Music October 2021 - January 2023 | Berlin, DE

Services and tools to distribute music, enrich metadata, process royalties, manage releases, monitor and grow music consumption

- Co-founded the MLOps function with the tech lead; built a Python framework for the end-to-end ML lifecycle (training, evaluation, serving) via declarative configuration, adopted enterprise-wide by data science teams
- Built model deployment infrastructure from scratch using BentoML & Yatai on GKE, serving multiple data science teams
- Designed and implemented a tool-agnostic experiment tracking library with declarative logging; deployed MLflow Tracking on GCP using Kubernetes, Helm, Terraform/Terragrunt, and Terratest (Go)
- Led knowledge sharing on Clean Code, SOLID, TDD, BDD, and IaC through workshops, code reviews, and mentoring across engineering levels

Machine Learning Engineer, Musimap SA (acquired by Utopia Music) ↗ June 2021 - September 2021 | Berlin, DE

Musimap adds value to your music catalogs and delivers knowledge on your music listeners

- Designed and implemented core components of Utopia Discover, a music recommendation product: REST APIs, Approximate Nearest Neighbor search, user interaction store, pre/post-processing services, and internal Python libraries for model training, evaluation, and serving

Data Scientist, Loudly GmbH ↗ May 2020 - May 2021 | Berlin, DE

Create, customize and discover music with the power of AI

- Improved music generation and recommendation systems behind Loudly AI Music Studio by refining GAN architectures for audio inpainting, building music similarity embeddings, and developing an auto-tagging system
- Rebuilt the ETL pipeline from scratch, improving training data quality and reliability

TECHNICAL SKILLS

Languages: Python, Rust, Bash
Concurrency: asyncio, threading, multiprocessing
ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, XGBoost, Hugging Face, ONNX
LLM Serving: vLLM, continuous batching, token streaming
Model Serving: BentoML, Yatai, FastAPI, Arrow Flight RPC, gRPC, REST
Inference Optimization: ONNX Runtime, Ampere, OpenVINO (Intel), Qualcomm AI
Infrastructure: Docker, Kubernetes, Terraform, Terragrunt, Pulumi
Cloud: AWS (EKS), GCP (GKE)
Observability: OpenTelemetry, Prometheus, Grafana, Sentry, PostHog, data drift monitoring, alerting
Experiment Tracking: MLflow, Comet
Databases: PostgreSQL, SQLAlchemy
Practices: TDD, BDD, DDD, Scrum, Clean Code, SOLID, IaC
CI/CD: GitHub, GitLab

EDUCATION

AI Coding Crash Course	Matt Pocock, aihero.dev (2026)
Software Architect Mindset	ArjanCodes (April 2026)
Software Designer Mindset	ArjanCodes (March 2026)
Machine Learning Engineering for Production (MLOps)	Coursera (2022)
Electrical & Computer Engineering, BSc & MSc	University of Patras (2012 - 2019)